

they would need to tackle projects that might have seemed unthinkable until then.

Among other things, the humble controller board provided the spark that ignited the 3D printing revolution. Although the first 3D printers were developed in the 1980s, these were still costly and relatively rare, typically found only in large corporate R&D labs. This changed in 2005 when a team at University of Bath in the UK began a project to develop a low-cost 3D printer that could print most of its own components. The team named the project RepRap, which is short for replicating rapid prototyper. The machine would melt inexpensive plastic filament and extrude it through a print head that would be moved across X, Y and Z axes using inexpensive stepper motors. Each of these motors was controlled by its own Arduino board, while other Arduinos took care of various housekeeping functions.

An early RepRap prototype printed its first useable part around a year later, paving the way for a subsequent design that could print more than half of its functional parts in February 2008. Tech enthusiasts from around the world quickly embraced RepRap project, using the open source design to build their own machines and contribute improvements that they had made. Soon, entrepreneurs were using RepRap's genetic code as the basis for affordable commercial products, such as Thing-o-Matic, a sub-US\$ 1000 3D printer kit, produced by MakerBot. Thing-o-Matic and virtually all other early machines used Arduino controllers. In addition, most low-cost units sold today use Arduino-style controllers.

Arduino also played a seminal role in the development of low-cost multi-rotor and fixed-wing drones that enjoy wild popularity with both hobbyists and commercial operators. Much of the early activity of this revolution occurred, and is documented on DIY Drones website called <https://diydrones.com>, an online forum for hobbyists interested in developing autonomous model aircraft.

Phase 3: Open source silicon. Although, open source technologies gave developers royalty-free access

to hardware and software platforms that they could modify to suit their needs, integrated circuits (ICs) their platforms run on have been almost exclusively based on proprietary IP. This began to change as devices known as field-programmable gate arrays (FPGAs) became inexpensive enough to be used as custom ICs. Designers could program the devices with desired functionality using large libraries of FPGA code (both proprietary and open source) in a matter of days or weeks, as opposed to months and millions of dollars required to develop the tooling for real ICs.

Although, FPGAs are excellent solutions for certain applications, these are slower, more power hungry and significantly more expensive than equivalent single-purpose silicon devices. Until recently, nearly all hard silicon designs were based on proprietary IP, either developed in-house or licensed from a third-party vendor.

Cost and time required to develop a reliable, high-performance processor architecture meant that nearly all ICs that require a processor core use third-party IP, licensed from a handful of vendors such as ARC, ARM, MIPS and Qualcomm. These cores offer excellent technical support, optional architectural enhancements and deep libraries of application code, but the IP tends to be very costly to license and usually has significant restrictions on how it can be modified.

As a result, the same market pressures that created open source software and hardware are giving rise to the first generation of open source silicon projects. Among the most ambitious and successful of these efforts is RISC-V project, an open instruction set architecture (ISA), originally developed at University of California at Berkeley, and currently supported by RISC-V Foundation.

Like most modern ISAs, it is based on reduced instruction set computing (RISC) principles that provide a good balance between silicon footprint, execution speed and code efficiency. The instruction set can be instantiated with a 32-, 64- or 128-bit word width, and has been designed to efficiently support a wide variety of common applications. Unlike proprietary ISAs, RISC-V

ISA can be freely used for any purpose, enabling anyone to design, produce or sell RISC-V chips and software.

RISC-V Foundation's 100+ member organisations have pooled their knowledge, know-how and resources to create a common platform and an ecosystem of development tools that form the basis of their own unique and highly-competitive products. SiFive, for example, is developing a service that allows its customers to design custom RISC-V-based system on a chip (SoC) devices and receive sample chips within a matter of weeks. SiFive also enables newcomers to become familiar with the technology by offering a low-cost, Arduino-compatible board powered by one of its RISC-V processors.

RISC-V's customisable architecture makes it easy for developers to create devices with the precise mix of functionality, performance and energy-saving features that give their products the unique capabilities they need to dominate a particular application space. As an example, Trinamic, a German semiconductor maker focussed on motor drivers, is adopting the RISC-V as the foundation for its next generation of motor drivers and power electronics.

The amount of computational power required for motor control functions has been steadily increasing to meet faster response times, higher precision and greater overall performance required by many emerging applications. RISC-V architecture makes it easy to add cores to driver ICs that are responsible for communication and protocols, offering more complete solutions and higher levels of integration.

The road ahead

As RISC-V and other early open source silicon platforms emerge, their development ecosystems will get a head-start from the rich trove of resources already created by open software and hardware movements. By enabling small, agile groups of entrepreneurs to bring cutting-edge products to market quickly and in a cost-effective manner, these open source technologies will continue to democratise and accelerate innovation within the tech economy. **EFY**

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